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Concordant late Paleozoic paleomagnetizations from the Atacama Desert: implications for tectonic models of the Chilean Andes

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Paleomagnetic results are reported from three formations of late Paleozoic age from the northern Chilean Andes of the Atacama Desert. For the first time primary NRM components are resolved for Paleozoic units along the western flank of the central Andes. Pole positions are calculated for the formations, and compared with APW data for cratonic South America. These comparisons reveal that the collecting sites in the northern Domeyko and Almeida Ranges of the central Andes have undergone no paleomagnetically defined rotations or translation with respect to cratonic South America since the time of NRM acquisition, which is likely to have been in the lower parts of the Kiaman Reverse Interval. If growth of the South American lithosphere has involved accretion of exotic microplates they are either likely to be substantially older than units sampled here, or be restricted to more coastal terranes. The results, taken together with other paleomagnetic data from northern Chile and southern Peru which have showed a wide range of discordance in their declinations when compared to each other or APW data, lead to the conclusion that this region of the Andes during the Mesozoic or Cenozoic has not been affected by simple processes of clockwise oroclinal bending from Peru to Chile, nor regionally consistent patterns of block rotations.

1. Introduction

In recent years a body of paleomagnetic data has begun to accrue from upper Paleozoic and younger rocks within the central Andean orogenic belt. A number of results from Peru and northern Chile, essentially around the Peru-Chile bend, have yielded discordant results when compared with the published data from cratonic South America [1–4]. A comparison of poles determined from orogenic localities with a South America APW path [5] is shown in Fig. 1. This serves to illustrate recently proposed hypotheses that the coastal Peruvian Andes (and perhaps northernmost Chilean Andes) may have undergone a counterclockwise rotation with respect to cratonic South America [1–3]. Data from further south in Chile, indicate that portions of the Chilean Andes have undergone clockwise rotations [4,6]. However most data from Chile have shown concordant inclinations. The discordances in declination from coastal regions have leant some support to Carey's theory [7] of oroclinal bending from Peru to Chile (Arica orocline) [1,2]. The data could also be used to suggest other mechanisms of rotation, such as localized, but systematic, rotations of coastal terranes in response to oblique plate convergence [6]. Alternatively, since many of the studies have failed to unambiguously rule out secondary magnetizations, some of the reported discordance may be artificial. Support that the suggested rotations in Peru may be real comes from the regional con-

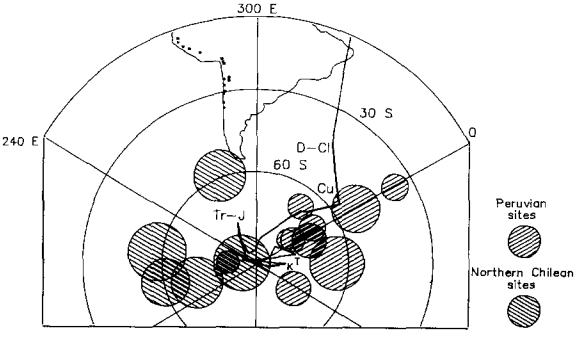


Fig. 1. Published paleomagnetic poles for the western Andean ranges in the vicinity of the Arica bend [1–4,24–26]. Also shown is a polar wander path for the Carboniferous to the Cretaceous [5].

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sistency of the counterclockwise discordance reported from practically all sites. Data from Chile does not show the same degree of consistency. While some results have shown clockwise discordance, some are concordant, and others are discordant in a counterclockwise sense.

In addition to the problem of understanding the nature and cause of internal rotations within the Andean orogenic system paleomagnetic data helps to isolate exotic microplates that may or may not have been incorporated into the Andean system. Thus far in Chile paleomagnetic results from only one coastal region (31°S) has revealed signs of paleolatitudinal discordance [4]. These preliminary results were from Triassic volcanic units (Pichidangui Formation) and suggested an equatorial drift of 10–15° for the sampling localities when compared to data from cratonic South America. These Triassic units are cut by Cretaceous plutons and dikes, which have also been the subject of paleomagnetic investigations. The results from the Cretaceous intrusions are basically concordant with South American APW path data [8] without any corrections for regional tilting. Thus if the discordance from the Triassic units is real, the representative coastal "terrane" is likely to have been incorporated into the Andean orogenic system prior to the Cretaceous.

Here we report new results from upper Paleozoic formations from the eastern portions of the Atacama Desert of northern Chile to further constrain the nature of displacements and rotations that may have occurred in the Mesozoic and Cenozoic within the Andes. These data come from the high western flank of the main cordillera, and help to resolve the extent of discordances along the western flanks of the central Andes that has thus far been principally documented for coastal regions.

In looking for displaced and rotated terrains by paleomagnetic means we concentrated on late Paleozoic units likely to have primary magnetizations and straightforward structural corrections. With these criteria in mind, three formations were sampled in separate fault bounded blocks to the south of the Salar de Atacama (Fig. 2a). Andesitic and rhyolitic flow units of the La Tabla Formation were sampled in Sierra de Argomedo (24°30′S, 69°20′W) along the western flanks of the Cordillera de Domeyko. Further to the northeast 2 localities (24°25′S, 68°33′W; and 24°21′S, 68°32′W) in the Pular and Cas Formations were sampled from Sierra Almeida. For the La Tabla and Pular Formations, sampling included volcanic conglomerate clasts from overlying units to help constrain the origin and age of magnetic remanence.

2. Geology

2.1. Sierra de Argomedo

This is an uplifted, fault bounded range of principally pre-Cretaceous units, that forms the western portions of the Cordillera de Domeyko to the west of the Salar de Punta Negra (Fig. 2b). The oldest unit is a marine clastic sequence which based on regional correlations is probably Devonian to early Carboniferous in age. Some disputed trace fossils have been used to indicate an early Paleozoic age [9]. These are unconformably overlain by acidic volcanic units of the La Tabla Formation that have been ascribed to an upper Paleozoic/lower Triassic magmatic belt that extended from south-central Argentina to northern Chile [10,11].

The units of the La Tabla Formation were sampled for this study on the western limb of an upright syncline along the eastern edge of the Sierra de Argomedo (Fig. 2b). Volcanic units here consisted primarily of rhyolitic flows and ignimbrites along with some porphyritic andesites and dacites. The La Tabla Formation is overlain by Triassic continental and marine clastic sequences, which are in turn overlain by Jurassic limestones [12]. In the main range of the Cordillera Domeyko, directly east of the Sierra de Argomedo range, polymict volcanic breccia and conglomerate beds were sampled from the basal portions of the Triassic sequence where they rest conformably over the La Tabla Formation. Given the lack of flows in the sequence, rounded clasts of volcanic rocks are presumed to have been derived from the underlying La Tabla Formation lavas, and were sampled for a conglomerate test.

2.2 Sierra de Almeida

The Sierra de Almeida range (Fig. 2c) is a fault bounded block east of the Cordillera de Domeyko and to the south of the Salar de Atacama. While ages of units in the range span much of the

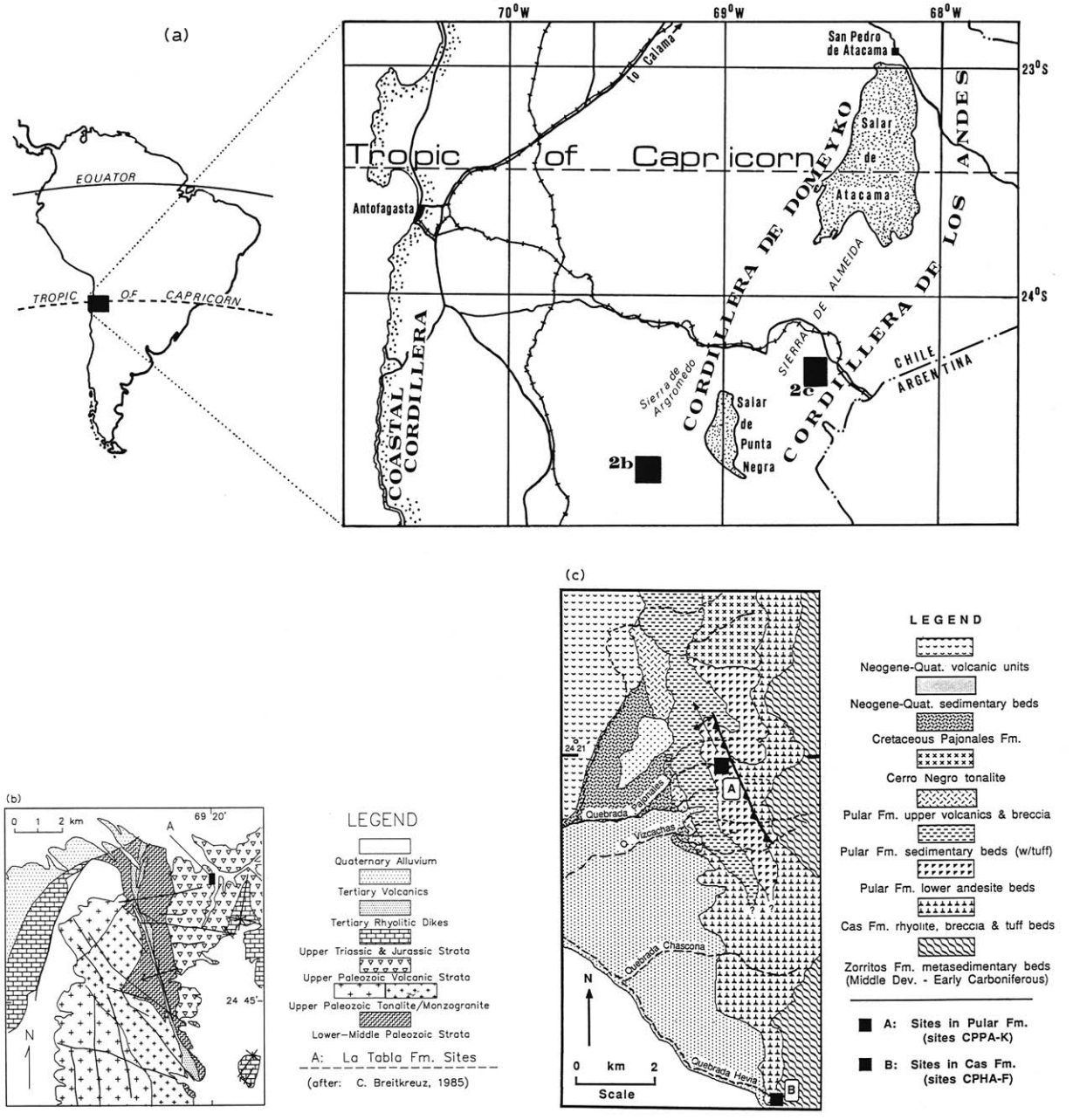


Fig. 2. (a) Location map of South America with the two collecting locations shown. (b) Simplified geologic map of the Sierra de Argomedo ranges in the Cordillera de Domeyko, west of the Salar de Punta Negra [9]. Sampling location is shown by black square. (c) Simplified geologic map of the collecting region in Sierra de Almeida [14]. Sampling sites are shown.

Phanerozoic, our collecting was restricted to the Cas and Pular Formations, which comprise a late Paleozoic sequence of volcanic and sedimentary rocks. At Quebrada Pajonales the two formations unconformably overlie the Devonian/Early Carboniferous Zorritas Formation, a shallow water marine clastic unit [11,13]. They are unconformably covered by the Cretaceous Pajonales Formation [11], a continental sequence of redbeds and conglomerates. The late Paleozoic sequence is ap-

proximately 2500 m thick. In Fig. 2c it is broken down into four lithostratigraphic units: the lowest most unit is presently mapped as the Cas Formation and is dominantly formed by massive rhyolitic flows. The upper three units are mapped as the Pular Formation which has a lower, dominantly andesitic, lava sequence, a central unit of continental, and probably marine, sedimentary beds, and an upper volcanic unit of principally andesitic lavas. The central sedimentary units carry ostracods of the Paraparchitidae family which suggest an age no younger than Carboniferous/Permian [14]. Five kilometers to the south of the Quebrada Pajonales (in Quebrada Hevia) the Pular Formation interfingers with, and dies out within, the rhyolytic flows of the Cas Formation. A biotite K-Ar age of 290 ± 7 Ma has been obtained from a biotite crystal tuff in the lower part of the central sedimentary unit that was collected just above the paleomagnetic sampling sites in Quebrada Hevia (Fig. 2c).

Sampling of the Pular Formation was restricted to the lower andesitic unit for sites used to calculate mean poles for the formation, and to a volcaniclastic conglomerate bed at the base of the central sedimentary unit for a conglomerate test. Volcanic beds were sampled along the western limb of an upright asymmetric anticline at Quebrada Pajonales, while the rhyolitic flows of the Cas Formation were sampled further south in Quebrada Hevia. Because the Pular Formation interfingers and dies out with the rhyolitic lavas of the Cas Formation north of the sampling sites in the Quebrada Hevia area, we cannot be certain of the relative stratigraphic position of the two sampling sites other than to say that they are likely to be fairly close in stratigraphic position.

3. Paleomagnetic techniques

In the field, cores 2.54 cm in diameter were drilled using either a gas or electrically powered drill. Each sample was then oriented with an aluminium orienting devise to which a Brunton compass was attached. In the laboratory one 2.2 cm long sample was cut from each core for magnetic measurements. Tilt corrections were determined on the basis of the dip of the volcanic sequence in the vicinity of each site. A simple correction of rotating about the strike of bedding was used to bring beds back to horizontal since significant plunges of fold structures were not apparent at the sampling sites. Tilt corrections are provided in Table 1.

All magnetic measurements were made on a flux-gate spinner magnetometer [15] at Rutgers University. Alternating field (AF) demagnetizations were done on a Schonstedt GSD-1 AC Geophysical Specimen Demagnetizer and thermal demagnetization was done using a Schonstedt TSD-1 Thermal Specimen Demagnetizer.

La Tabla Formation (flows). Thirty cores, three per flow, were taken from 10 different flows of the La Tabla Formation. Natural remanent magnetizations (NRM) from these samples ranged in intensity from 0.3 to 130×10^{-3} A/m with the majority of samples below 5×10^{-3} A/m.

From each unit a pilot sample was chosen and subjected to either AF (5–90 mT in 11 steps) or thermal (100° to 600°C in 9–11 steps). AF demagnetization yielded more spurious results than thermal demagnetization for samples in the lower range of NRM intensities. Thermal demagnetization gave consistent and univectorial results over the entire spectrum of sample intensities (Fig. 3a).

Zijderveld and decay curves of these pilot samples generally show an unblocking temperature of about 500°C and Curie temperature of between 560° and 600°C. Based on this preliminary data magnetite of a likely primary igneous origin was suggested to be the carrier of remanence. However, because of the difficulties encountered with the low-intensity samples under AF demagnetization, all remaining samples were treated with thermal cleaning in a 4-step (400, 500, 560, and 600°C) procedure. On completion of the bulk thermal treatment only one sample out of thirty retained more than 10% of its original magnetization at 600°C, and almost all samples showed univectorial decay towards the origin. Fig. 3b shows the typical response of pilot samples to thermal demagnetization.

La Tabla Formation (conglomerate test). The basal conglomerate beds of the overlying Triassic units in the Cordillera de Domeyko are thought derived from the underlying La Tabla Formation. Due to problems encountered in coring at 14,000 ft only 14 cores of volcanic clasts were obtained in the

TABLE 1

Results from flow units

Site		N	k	R	α_{95} (°)	D (°C)	I (°)
Tabla Formation: 24°30′S, 290°40′E; all data tilt corrected—tilt correction = 010°, 53°							
PFTA		3	332.5	2.99	6.8	133.1	56.9
PFTB		3	535.3	3.00	5.3	133.7	59.7
PFTC		3	7.8	2.75	47.2	106.1	65.0
PFTD		3	105.9	2.98	12.0	147.3	57.6
PFTE		3	769.7	3.00	4.4	142.4	57.9
PFTF		3	968.4	3.00	4.0	149.0	59.6
PFTG		3	192.0	2.99	8.9	126.2	55.8
PFTH		3	26.7	2.93	24.3	115.2	65.9
PFTI		3	176.6	2.99	9.3	155.6	55.0
PFTJ		3	114.8	2.98	11.6	145.6	46.6
All sites		10	72.4	9.88	5.7	136.9	58.8
Cas Formation (Quebrada Hevia): 24°25′S, 291°27′E; all data tilt corrected—tilt correction = 230°, 30°							
CPHA		3	8.7	2.77	44.6	186.6	73.8
CPHB		3	648.9	3.00	4.8	139.8	38.3
CPHC		3	107.0	2.98	12.0	167.3	56.7
CPHD		3	35.3	2.94	21.1	169.2	50.1
CPHE		3	57.0	2.96	16.5	166.7	48.9
CPHF		3	167.9	2.99	9.5	131.4	43.9
All sites [a]		5	34.1	4.88	13.3	153.5	48.7
Hevia paleopole: 66°S, 2°W							
Pular Formation (Quebrada Pajonales): 24°21′S, 291°28′E; A component in situ; B component tilt correction = 015°, 35°							
CPPA	(A)	3	49.7	2.96	17.7	359.2	−56.0
	(B)		34.2	2.94	21.4	281.7	−49.7
CPPB	(A)	2	81.4	1.99	28.0	357.8	−45.6
	(B)		81.5	1.99	28.0	314.2	−66.9
CPPC	(A)	3	29.6	2.93	23.1	5.7	−25.8
	(B)		134.7	2.99	10.7	330.3	−59.4
CPPD	(A)	3	122.9	2.98	11.2	350.9	−36.0
	(B)		170.4	2.99	9.5	325.0	−60.1
CPPE	(A)	3	349.9	2.99	6.6	12.3	−38.9
CPPF	(B)	3	240.6	2.99	8.0	306.6	−68.7
All sites	(A)	5	39.7	4.90	12.3	1.4	−40.7
	(B)	5	42.2	4.91	11.9	310.3	−62.3
Pajonales paleopole: 45′S, 341°E							
Paleopole from La Tabla mean: 51°S, 347°E							
Paleopole from Pular/Cas means: 57°S, 350°E							

[a] Except CPHA, see text.

field. Of these only 12 yielded stable demagnetization trajectories. NRM intensities for these clasts ranged widely, from 10 to 9000×10^{-3} A/m. Compositions of the volcanic clasts are more variable than that represented by the samples taken from the lavas.

Pilot clasts were AF and thermally demagnetized in a manner similar to that described above. Typical Zijderveld plots are shown in Fig. 4. These plots seem to indicate a univectorial magnetite carrier, with a small portion of magnetization remaining after 90 mT in some samples. Complete thermal demagnetization were done on three clasts, two of which clearly show an unblocking temperature between 560° and 600°C. The third clast was rejected because its magnetization did not decay at all. In general AF demagnetization was carried out on the remaining clasts

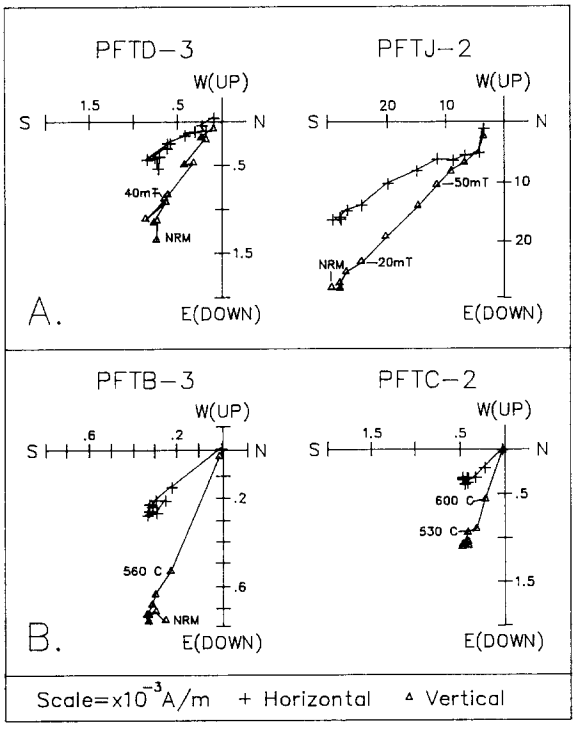


Fig. 3a. Four typical Zijderveld diagrams of AF and thermal demagnetization histories for samples from Sierra de Argomedo (A: AF demagnetization, B: thermal demagnetization).

until the sample appeared to be decaying in a straight line to the origin on a Zijderveld plot. At that point demagnetization was halted and that trajectory was taken to be the stable direction for that sample.

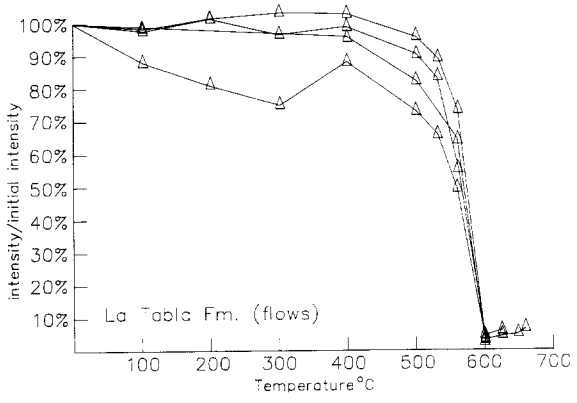


Fig. 3b. Plot of decay of intensity of remanent magnetization/initial intensity during thermal demagnetization for La Tabla Formation samples.

Pular Formation (volcanic units of Quebrada Pajonales). Eighteen cores were drilled in 6 flows of the lower andesitic units exposed in Quebrada Pajonales. NRM intensities varied between 50 and 300×10^{-3} A/m.

Pilot AF demagnetization of 4 samples revealed that samples were not univectorial, and likely to have two or more components. However, AF demagnetization was incapable of resolving their independent directions. Pilot thermal demagnetization proved more useful. In stereoplots, directions of magnetic remanence at successive stages of thermal demagnetization described a great circle path from the NE to the NW quadrant where directions stabilized for the remaining demagnetization steps. Given the multicomponent nature of these pilot samples, all remaining samples were treated with a 10 step (100, 150, 200, 300, 400, 500, 550, 600, 625, 650 °C) thermal demagnetization procedure. The typical demagnetization character of these samples is shown in Fig. 5a and b. Here a low-temperature (A) component appears in the first four steps (100–300 °C), while the high-temperature (B) component, is clearly seen in steps 5–10 (400–650 °C) as a linear component decaying towards the origin. Of the six flows, both components had well represented linear trajectories in four, and the other two had cores with good trajectories for either the A or B components.

Pular Formation (conglomerate test). The NRM intensities of the 18 clasts drilled at Quebrada Pajonales ranged from 50 to over 2000×10^{-3} A/m. The clasts were of three lithologic types; andesite, rhyolite, and welded ignimbrite. The rhyolitic clasts may have been derived from the underlying Cas Formation as well.

Pilot cores from each lithologic group were subjected to both AF and thermal demagnetization. A variety of behaviors was seen during demagnetizations, but the majority of the clasts proved to be univectorial over a wide range of temperatures and coercivities (Fig. 6). Only one clast showed an indication of having the low temperature component seen in the andesites.

Cas Formation (volcanic units of Quebrada Hevia). Fig. 7a shows some typical curves for pilot samples from Quebrada Hevia where 20 cores were taken from 6 flows. NRM intensities at these sites

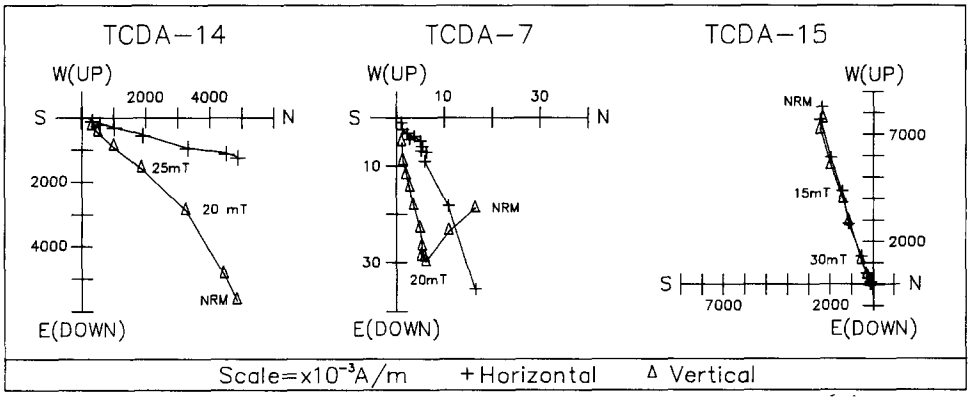


Fig. 4. Zijderveld diagrams of clasts used for the La Tabla Formation conglomerate test.

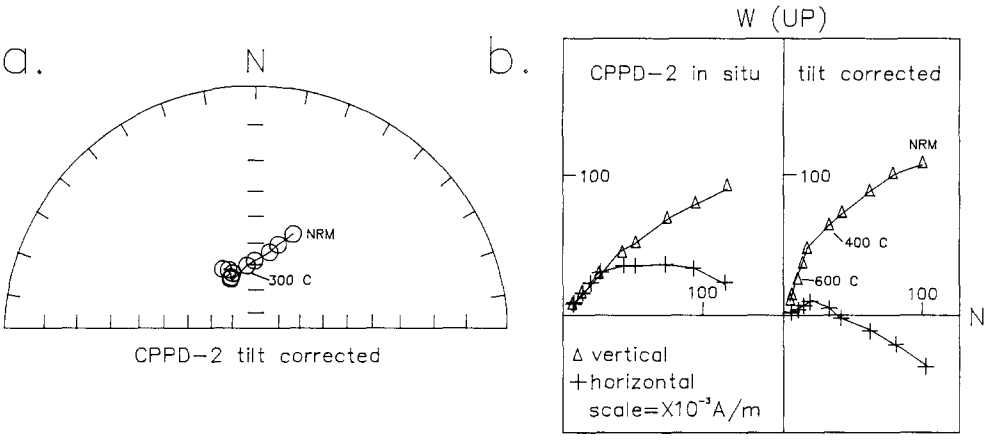


Fig. 5. Diagrams illustrating demagnetization characteristics for two component samples in Quebrada Pajonales (Pular Formation). At left is a stereogram of NRM of 650 ° C tilt-corrected directions plotted on the upper hemisphere. At right are tilt-corrected, and in situ, Zijderveld diagrams for the same sample.

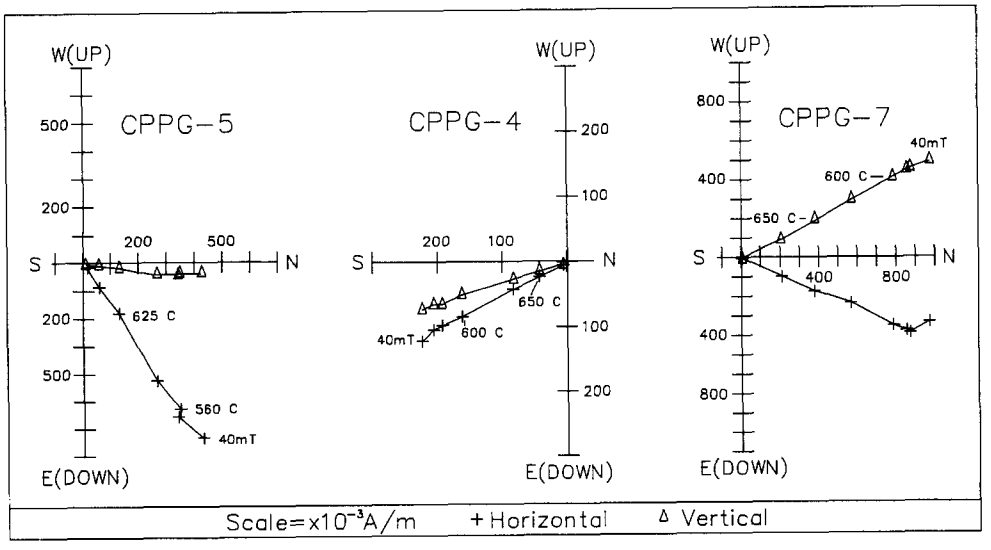


Fig. 6. Typical Zijderveld diagrams of clasts used for the Pular Formation conglomerate test.

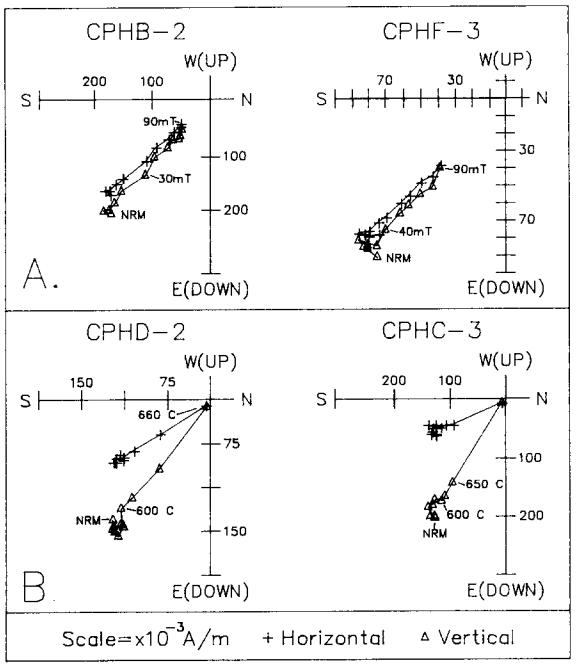


Fig. 7a. Four Zijderveld diagrams illustrating characteristic demagnetization histories for samples from Quebrada Hevia (Cas Formation, A: AF demagnetization, B: thermal demagnetization).

were generally between 100 and 200 $\times 10^{-3}$ A/m.

After complete demagnetization and analysis of pilot samples, a 4-step (400, 550, 610, 660 °C) bulk thermal demagnetization procedure was again chosen to clean up the remaining cores. Thermal decay curves indicate complete removal of remanence between 660 ° and 670 °C. Both AF and thermal treatment show that these rocks have a number of behaviors during demagnetization. This is particularly apparent in the AF decay curves (Fig. 7a, b). Despite the possibility of more than one carrier of magnetic remanence, Zijderveld plots for these samples indicate that the remanence is univectorial throughout the sample population over a wide range of temperatures and apparent sample coercivities.

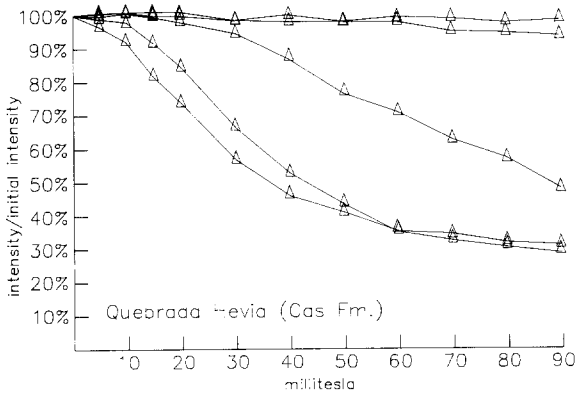


Fig. 7b. Plot of the decay of intensity of remanent magnetization/initial intensity during thermal demagnetization for Cas Formation samples from Quebrada Hevia.

Component analysis of statistics. All data manipulation and plotting were done on a microcomputer. Initial pilot samples were visually inspected for their decay characteristics. Following the complete cleaning of cores from all sites, each sample was subjected to a least-squares best fit analysis [16] of a minimum of four demagnetization steps. The directions obtained from three independently oriented cores from each flow were then averaged to yield a mean site (flow) direction according to standard paleomagnetic techniques [17,18]. At each collecting locality site means were again averaged to yield an overall mean declination and inclination for the locality. In addition, sites from Quebrada's Hevia and Pajonales were grouped together to obtain a mean declination and inclination, for the late Paleozoic sequences of Sierra de Almeida before calculation of a composite paleopole.

For the conglomerate tests, each core (clast) was treated to component analysis, and then results from all clasts of a given locality grouped to calculate the Fisher statistical parameters and the R parameter for randomness [19].

4. Results

4.1. La Tabla Formation

The mean directions obtained for the La Tabla Formation are believed to be representative of a primary magnetization. The carrier is most likely magnetite of a primary igneous origin. This is inferred from the positive conglomerate test, and the response of the samples under AF and thermal demagnetization.

The conglomerate test for the La Tabla Formation was reduced to the statistical analysis of 12 cores after two were rejected for lack of any well defined stable magnetic components. The results

are shown in Table 2. The mean unit vector directions for all clasts were vectorially summed to yield a mean vector of length R. This length, when compared to the number of unit vectors in the population, can be used as an indication of randomness. For a population of 12, R values less than 5.52 are considered random [18]. The value for this conglomerate test is 4.15. While the population is small, univectorial components were carefully isolated for this test. The composition of the volcanic clasts were more variable than that of the sampled lavas, perhaps explaining their greater range of intensities. They did, however, include clasts of similar composition to the volcanic units sampled from the La Tabla Formation, and under demagnetization many show similar coercivities and blocking temperature spectrum.

Both the samples from La Tabla Formation flows and volcanic clasts from the overlying conglomerate beds, appear to have magnetite as the carrier of remanent magnetization. Typical of magnetite all these igneous samples showed decay under AF treatment, and have Curie temperatures between 560° and 600° C. This magnetite carrier is earlier than the deposition of the conglomerate beds, and probably of primary igneous origin in the volcanic flows.

Proceeding with the assumption that the magnetizations measured in the flows from the La Tabla Formation are primary, the results shown in Table 1 for the 10 flows are interpreted to represent a reversed magnetization with a tilt corrected mean declination of 136.9°, a mean inclination of 58.8°, and an α_{95} of 5.7°. This yields a paleopole with lat. = −51° S, and a long. = 347° E. The age of the La Tabla Formation on regional grounds is believed to be post Early Carboniferous, but pre-Late Triassic.

4.2. Pular and Cas Formations

The results from Quebrada Hevia show a magnetization, that is stable during both AF demagnetization and thermal demagnetization to a temperature of approximately 660° C. The stable univectorial decay both above and below the Curie temperature of magnetite leads us to believe that we are seeing a univectorial magnetization of a reverse direction that is carried probably by both magnetite and hematite. The decay of remanence under AF treatment is uncharacteristic of hematite, and typical of magnetite, yet the persistence of remanence above the Curie temperature of magnetite suggests the presence of hematite. While the magnetite carrier would likely be of a primary igneous origin, the hematite carrier would have to have originated as an oxidation product within the volcanic units. For reasons discussed further on, this oxidation is not likely to have been younger than middle Permian. The results for the Quebrada Hevia locality are shown in Table 1. Unfortunately bed CPHA could not be used due to an ambiguity that arose subsequent to field work over orientations of fragmented cores. The tilt corrected mean for the 5 remaining sites is $D = 153.5°$, $I = 48.7°$, with an α_{95} of 13.3°.

Two stable components of magnetization were isolated from samples taken from the Pular Formation in Quebrada Pajonales. Five sites carried both the A and B components. The low-temperature A component was found to be a northerly and up (normal) magnetization that remained stable up to maximum temperature of approximately 400° C. The in situ mean for the five sites containing the A component is $D = 1.4°$, $I = -40.7°$, with an α_{95} of 12.3°. When transformed into a pole position this in situ direction coincides extremely well with the position of the present-day geocentric dipole. The five sites containing the high-temperature component yielded a northwesterly and up (normal) orientation with an overall (tilt-corrected) mean of $D = 310.3°$, $I = -62.3°$, and an α_{95} of 11.9°. Significantly, this direction is

TABLE 2

Results of conglomerate tests [a]

Site	N	k	R	α_{95} (°)	D (°)	I (°)
La Tabla	12	1.4	4.15	66.0	13.9	29.9
Pular	17	1.2	3.37	80.4	6.2	53.4

[a] Tilt correction for La Table Fm. conglomerate test = 010°, 53°
Tilt correction for Pular Fm. conglomerate test = 015°, −35°

close to being antipodal to the directions of the Quebrada Hevia sites. The in situ coincidence of the A component with a present-day field direction, and the approximately antipodal character of the tilt-corrected B component to the tilt-corrected Quebrada Hevia sites argue that we are seeing a recent secondary low-temperature component overprinting an early (pre-tilt) high-temperature component. However, like the samples from Quebrada Hevia, the high-temperature component appears to be at least in part carried by hematite. Further evidence of this can be seen from the two anomalous curves (CPPF-1, CPPF-3) in Fig. 8. The two samples for these curves were from very red, highly oxidized, flows and showed no evidence of the low-temperature component. Thus, while the carrier of remanence for the low-temperature component is unknown, the high-temperature component appears, as in Quebrada Hevia, to be a univectorial combination of a TRM and a rapidly acquired CRM. The hematite component is very possibly a high temperature oxidation product produced during initial cooling of the lavas [20].

The results of the conglomerate test at Quebrada Pajonales are listed in Table 2. Unlike the underlying andesitic flows, the samples are dominantly univectorial, with the exception of one clast, which may have possessed the low-temperature A component. After appropriate cleaning and component analysis, these samples are used, like in the other conglomerate test, to calculate an R value.

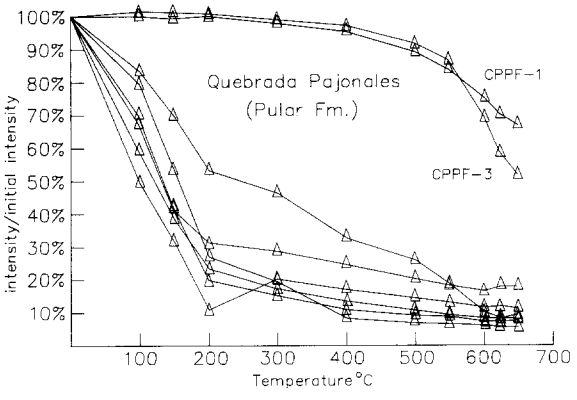


Fig. 8. Plot of the decay of intensity of remanent magnetization/initial intensity during thermal demagnetization for samples from sites in Quebrada Pajonales (Pular Formation) showing anomalous curves for samples at the CPPF site.

For the population of 17 clasts a value less than 6.60 indicates randomness, and for these samples $R = 3.37$.

The primary or early nature of the remanent magnetizations seen in the volcanic units of both Quebrada Pajonales and Hevia samples is supported by indications of a magnetite carrier, as well as the positive conglomerate test. However, as previously noted, the normal magnetizations seen in the samples from Quebrada Pajonales are close to being antipodal to the reverse magnetizations seen in the units from Quebrada Hevia. Thus there are three independent lines of evidence to suggest that the magnetizations acquired here are early TRM (for magnetite) and CRM (for hematite) components. This is an argument supported by the fact that the only other component isolated in the samples turns out to be a low-temperature component that in situ coincides with present-day field directions.

Combining the normal directions from Quebrada Pajonales (5 flows) with the reverse directions from Quebrada Hevia (5 flows) yields a mean declination of 143.8° and inclination of 56.1° ($\alpha_{95} = 9.6°$). The paleopole calculated from this data has a position of lat. = 57.0°S and long. = 350°E.

5. Discussion

5.1. Age constraints

The age constraints for the La Tabla Formation is based, at present, only on regional correlations among Paleozoic and early Mesozoic units of the region. K/Ar age dating is in progress in Chile over these late Paleozoic volcanic sequences [21,22] and in the near future more rigorous constraints will be available. As discussed earlier regional correlations place an approximate age of Middle Carboniferous to pre-Triassic on the formation. The age of the Pular Formation and Cas Formations are much better constrained, both paleontologically and radiometrically. Fossils from the central marine units (directly overlying our sites) in Quebrada Pajonales, indicated that the Cas and Pular Formations are no younger than Carboniferous to Permian. In addition in the outcrops of the Cas Formation in Quebrada Hevia, only meters over the beds sampled for paleomagnetic work, a biotite K/Ar age of 290 ± 7 Ma has been ob-

tained from the biotite–crystal tuff. Thus a late Carboniferous age is indicated. The fact that paleomagnetizations are reversed in the Hevia sites under the dated tuff, would argue for magnetizations being acquired in the lower parts of the Kiaman Reverse Interval. There is some question, based on more recent mapping, as too exact stratigraphic correlations between the Quebrada Pajonales and Hevia sites. If the Pajonales sites are stratigraphically lower than the beds sampled in Quebrada Hevia, then they are likely to represent the normal event at the base of the Kiaman. Alternatively if they are younger, then we may be seeing one of the normal events within the Kiaman suggested from work not too distant in Argentina [23].

Finally, while perhaps slightly circular in reasoning, the coincidence in paleopoles determined from the Pular, Cas, and La Tabla Formations is supportive of their stratigraphic correlation, over and above the regional stratigraphic constraints and their lithologic similarities.

5.2. Comparisons to South America APW path

In Fig. 9 the results from both the Pular, Cas and La Tabla Formations are shown on a Carboniferous to 200 m.y. B.P. APW path for South America. This path was constructed by averaging individual pole determinations in 30 Ma windows to determine mean paleopoles every 10 m.y. [5]. The data from the Pular, Cas, and La Tabla Formations yield poles directly on the APW path for the Late Carboniferous to earliest Permian.

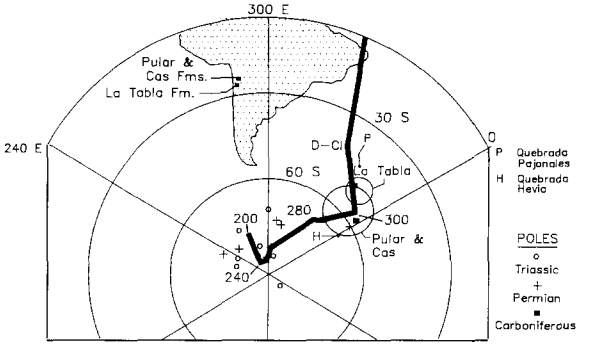


Fig. 9. Polar projection map showing an averaged APW path for South America [5] together with the poles calculated for the Pular, Cas and La Tabla Formations. Circles represent α_{95} confidence limits for the La Tabla Formation pole, and a pole combining the Cas and Pular Formation results. The two points labeled *P* and *H* are the individual poles for sites of Quebrada Pajonales (Pular Formation) and Quebrada Hevia (Cas Formation).

The concordance of the results determined here with the APW path is significant for three reasons. First, it argues that these portions of the Andes have been an integral part of South America since the Carboniferous. If terranes have been added to the Andes, they must lie outboard, or be significantly older. Second, it argues for little if any tectonic rotations in the sampling sites that can not be accommodated by simple corrections for tilt of bedding. While further mapping may refine these corrections, at least at present, it appears that the discordances reported from more coastal regions directly west and southwest, either represent regionally restricted rotations or are in part artificial. Large-scale clockwise rotation of the northern Chilean Andes is certainly not consistent with these data. Lastly, the general concordance of the results from the three formations to each other as well as to the late Carboniferous APW data, lends support to age assignments and stratigraphic correlations that have been made among late Paleozoic units of northern Chile. Along this line we note that ages of middle Permian or younger would result in discordance with the APW path. This is a conclusion also consistent with the fact that the majority of the samples (La Tabla and Cas Formations) are reversely magnetized. Given the potential for additional K/Ar dating of stratigraphically lower and higher volcanic units in the Pular and Cas Formations, as well as their thickness and abundance of volcanic units, a comprehensive magnetostratigraphic study of the formation would likely yield useful information on the Kiaman Interval, and further regional age constraints for the late Paleozoic evolution of this region of the Andes.

The discordant results obtained from other studies in the Andean cordillera, in both Peru and Chile, have rarely been reinforced by fold or conglomerate tests to constrain the age of magnetization. While reversals have been noted in some of the studies, the general stability of post-Paleozoic pole positions with respect to the present geocentric dipole is not very reassuring for use of reversed components as a field test for primary remanence. Also most of the coastal terranes have

a prolonged history of igneous activity, generally continuing into early Tertiary times. If, indeed, there is a broad counterclockwise bending of the Peruvian Andes then perhaps our attention should turn at the present time to the main Peruvian Cordillera, where volcanic units, equivalent to those studied here can be sampled inboard of post-Paleozoic igneous activity, and with good potential for field tests.

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